THE QUOTIENT OF A COMPLETE LINEARLY TOPOLOGIZED MODULE BY A CLOSED SUBMODULE IS COMPLETE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one family of short exact sequences; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Exactness of an inverse limit of short exact sequences whose transition maps are surjective. Algebra, Lemma algebra-lemma-ML-exact-sequence, invoked at proof lines 989-991. Gives the surjection from the completion of M onto the limit defining the quotient.
- The closure of a submodule is the intersection of its images in the quotients by a fundamental system of open submodules. Same chapter, Lemma lemma-closed, invoked at proof lines 992-994. Identifies the kernel of the map as N, using that N is closed.

Source reference. Original lemma 0598: sources/stacks-chow/tags.txt, source lines 6806–6806. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AMS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 922–922. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let R be a topological ring. Let M be a complete, linearly topologized R-module. Let $N \subset M$ be a closed submodule. If M has a countable fundamental system of neighbourhoods of 0, then M/N is complete and the map $M \to M/N$ is open.*

Proof. Let M_n, $n \in \mathbf{N}$ be a fundamental system of open submodules of M. We may assume $M_{n+1} \subset M_n$ for all n. The system $(M_n + N)/N$ is a fundamental

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system in M/N. Hence we have to show that $M/N = \lim M/(M_n + N)$. Consider the short exact sequences

$$0 \to N/N \cap M_n \to M/M_n \to M/(M_n + N) \to 0$$

Since the transition maps of the system $\{N/N \cap M_n\}$ are surjective we see that $M = \lim M/M_n$ (by completeness of M) surjects onto $\lim M/(M_n + N)$ by Algebra, Lemma 0598. As N is closed we see that the kernel of $M \to \lim M/(M_n + N)$ is N (see Lemma 0AMS). Finally, $M \to M/N$ is open by definition of the quotient topology. $\square$